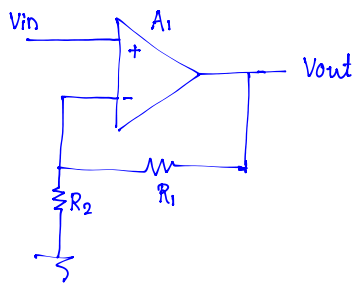
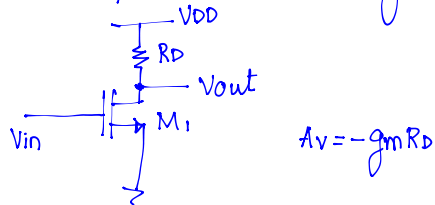


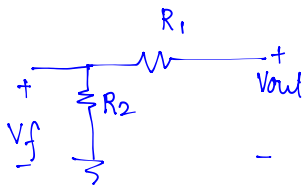
Operational amplifiers:-

1. Gain :- Amplifier to be designed with gain 10 and gain error less than 1%



$$A_{\text{closed}} = \frac{A_v}{1 + \beta A} = A_{\text{actual}}$$

$$A_{\text{ideal}} = \frac{1}{\beta} = \frac{R_1 + R_2}{R_2} = 1 + \frac{R_1}{R_2} = 10$$



$$V_f = \frac{R_2}{R_1 + R_2} V_{\text{out}}$$

$$\frac{V_f}{V_{\text{out}}} = \frac{R_2}{R_1 + R_2} = \beta$$

$$\text{Error} = \frac{A_{\text{ideal}} - A_{\text{actual}}}{A_{\text{ideal}}}$$

$$1\% = 0.01 \leq 10 - \frac{A}{1 + \frac{1}{10}A}$$

$$A \geq 990$$

$$A \geq 1000$$

$$\epsilon = \frac{1}{\beta} - \frac{A}{1 + \beta A} =$$

$$= \cancel{\beta} \times \frac{1 + \beta A - \beta A}{\cancel{\beta} (1 + \beta A)} = \frac{1}{1 + \beta A}$$

$$\epsilon = \frac{1}{\beta A}$$

$$0.01 \geq \frac{1}{\beta A}$$

$$0.01 \geq \frac{10}{A}$$

$$A \geq \frac{10}{0.01} = 1000$$